Program Elective-IV
User Experience Design (UX/UI)



Objective

1. Understand the basic concepts of user interface/user experience Design in order to design with intention
2. Achieve a deep understanding of the entire lifecycle of the design process
3. Provide a visual understanding of the product to make user interaction as easy and efficient as possible
4. Understand various design technologies for mobile and web to help avoid common mistakes and meet user requirements

Outcomes

1. Analyse the relationship between user interface and user experience to create user-centered designs
2. Apply various user experience research methods to improve user experience
3. Make use of various design principles to create intuitive wireframes and prototypes
4. Understand mobile and web-based user experience designs by integrating emerging interaction technologies

Examination Scheme

- T1 - 25 Marks
- T2 - 25 Marks
- ESE - 50 Marks
- Credits - 3

20 PECE 901 LB
User Experience Design Laboratory

Teaching Scheme :

- Lectures: 2 Hours / Week

Examination Scheme :

- In-Semester (TW): 25 Marks
- End Semester(OR): 25 Marks

Credits: 1

Objectives

To facilitate the learner to

1. Understand users' needs, experiences, behaviours and goals

2. Learn how visual perception affects the viewing experience

3. Explain Why you made design decisions, through presentations of assignments

Outcomes

After completion of the course, students will be able to

1. Explore techniques to understand user, their needs, values, abilities, and limitations

2. Design innovative, user-friendly interfaces for both mobile and/or web applications

3. Evaluate existing interface designs and redesign services and products to elevate the user experience

4. Utilize the industry-standard tools and deliverables essential to professional user interface design practices

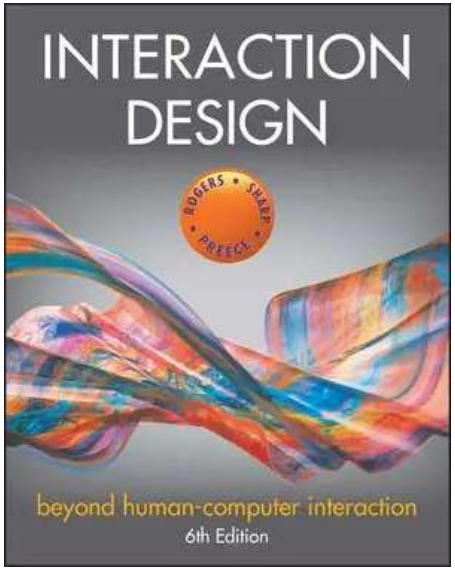
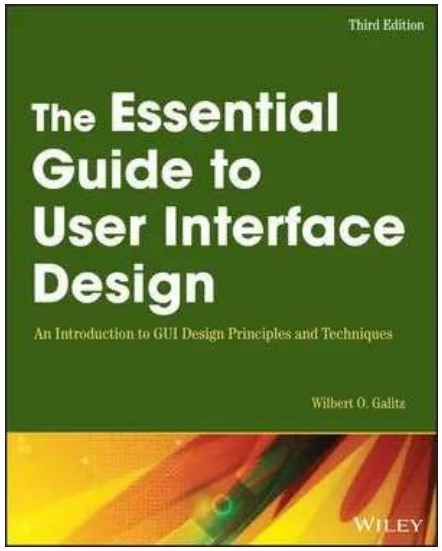
List of Assignments

1. Design user persona for the users of selected product / system and
2. Conduct a contextual inquiry for selected product / system.
3. Heuristic evaluation on a computer prototype developed by your classmates.
4. Design of User interface for the system using various interaction styles.
5. Design appropriate icons pertaining to a given domain. (Eg. Greeting cards)
6. Design a Mobile App/Website that can help people to sell their handmade products in metro cities
7. Improve Instagram with a new, innovative feature, which stands out from other image apps.
8. Redesign a page from the job portal you like (preferably a complex screen). Justify your selection and the changes/design you made. Document your design process on Notion.
9. ATM machine/KIOSK screen design for rural people
10. Tool exploration Adobe XD, Figma

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Text Books:

Interaction Design: Beyond Human-Computer Interaction: Book by Helen Sharp, Jenny Preece and Yvonne Rogers	Wilbert O. Galitz Wiley ‘The Essential Guide to User Interface Design’ 3rd Edition
	

Reference Books:	
1.	Don Norman, 'The Design of Everyday Things', Basic Books, A member of the Perseus Books Group, (2013)
2.	Ben Shneiderman, Catherine Plaisant, Maxine Cohen, Steven Jacobs, 'Designing the User Interface: Strategies for Effective Human-Computer Interaction', Pearson Education Limited (India),(2010)

Unit - I
Introduction to User Experience

Introduction to User Experience

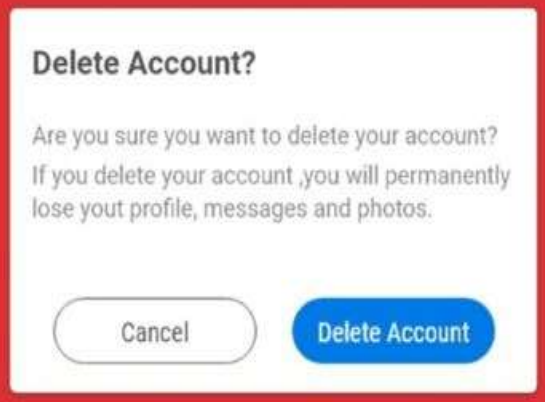
- What is User Experience
- Relationship Between UI and UX
- Why is UX Design so Important, What is UX Design and Where is Used
- Usability: A part of the User Experience, Understanding User Experience
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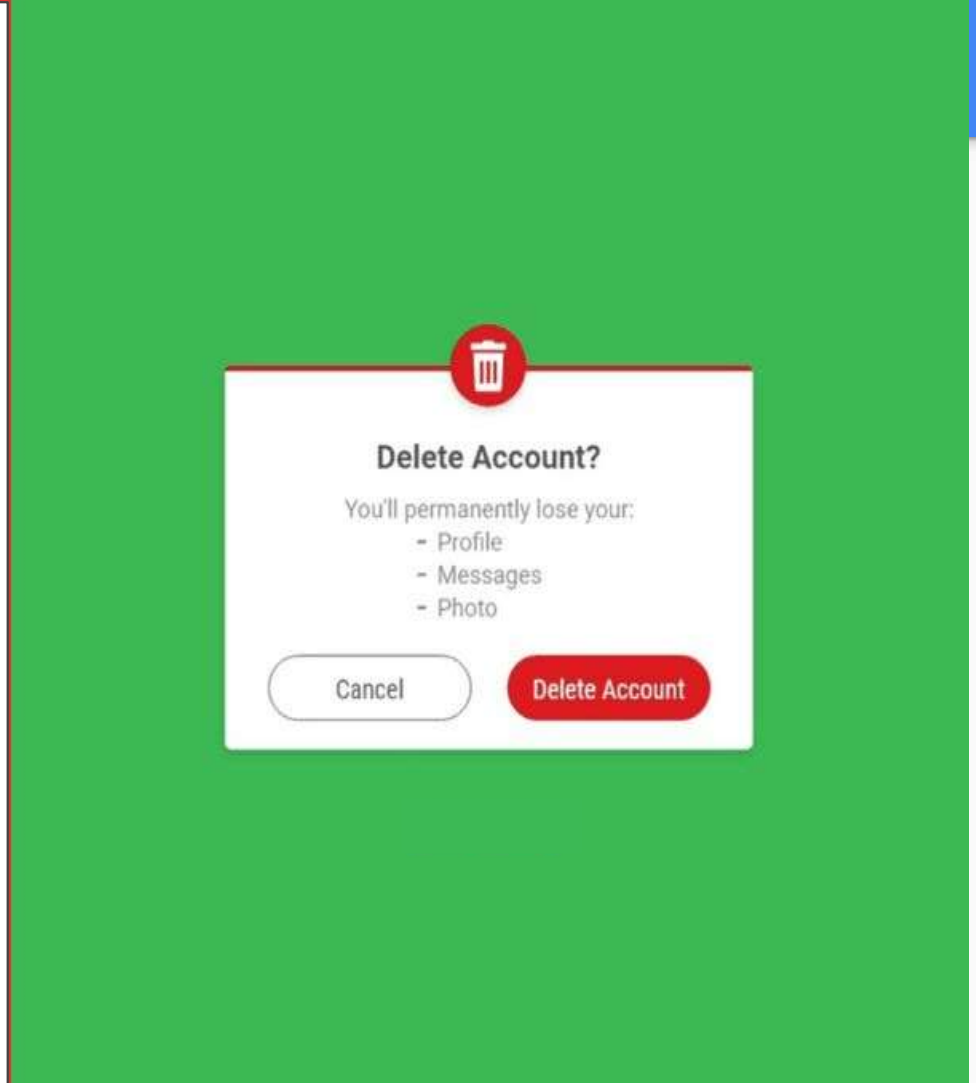
Introduction to User Experience

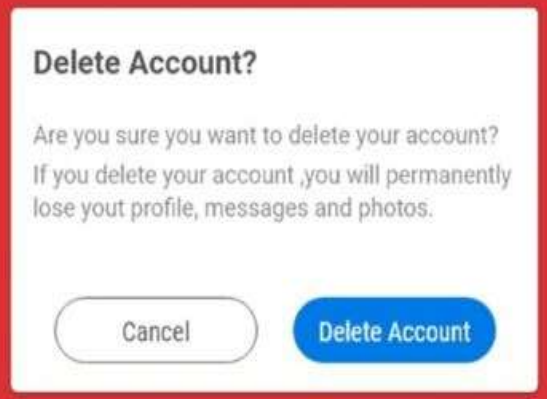
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What is User Experience

- The term '**user experience**' was coined by Dr. Donald Norman in the mid-1990's.
- He describe the importance of **user-centred design** – the notion that design decisions should be based on the **needs and wants of users**
- **User experience, or UX**, is a term used to describe the overall experience a user has when interacting with a product or service in a given context.
- Depending on how the product or service is designed, the experience can range from delightful to downright frustrating!.
- **UX is all about the user's interaction or experience with a product or service.**


Delete Account?
Are you sure you want to delete your account?
If you delete your account ,you will permanently lose yout profile, messages and photos.
Cancel
Delete Account


Delete Account?
You'll permanently lose your:
- Profile
- Messages
- Photo
Cancel
Delete Account



Bad UX!



Good UX!

Examples of Bad Designs





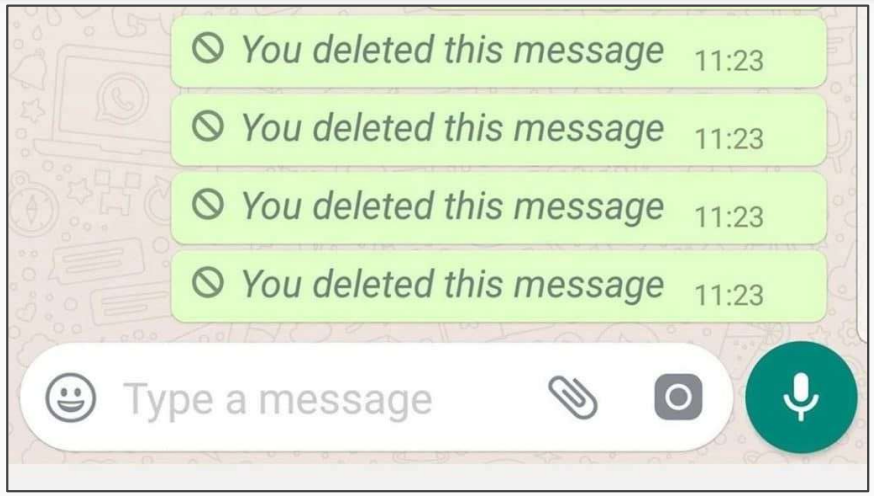
Apple Mouse

Examples of Bad Designs





Apple Mouse

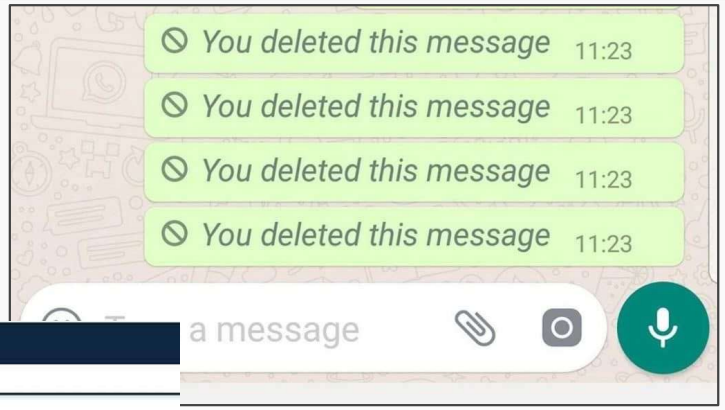


Examples of Bad Designs





Apple Mouse



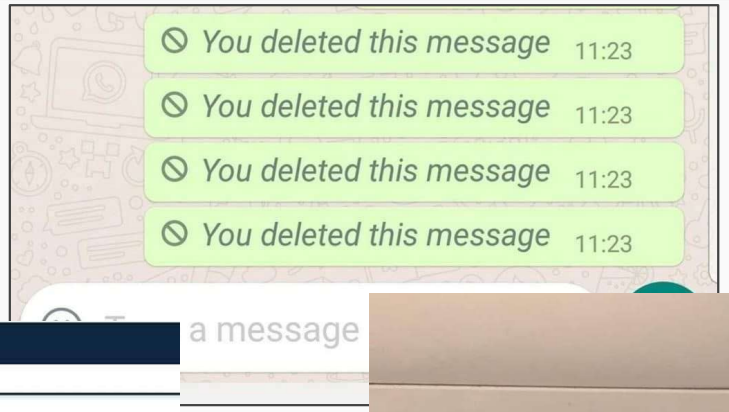


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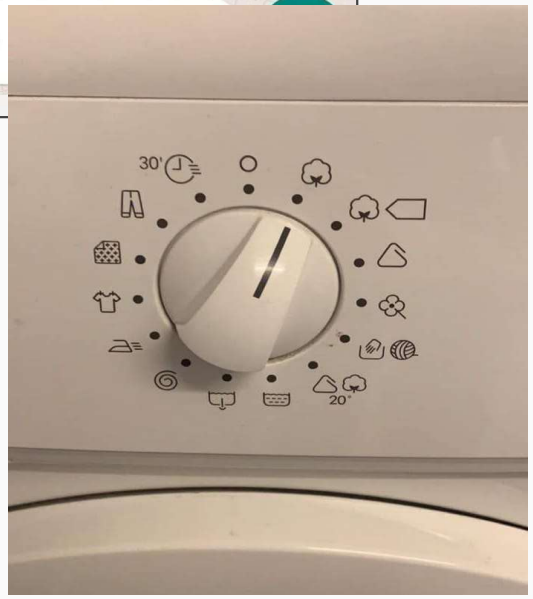




Apple Mouse







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What is User Interface

- The user interface is the point at which human users interact with a computer, website or application.
- The goal of effective UI is to make the **user's experience** easy and intuitive, requiring minimum effort on the user's part to receive the maximum desired outcome.
- UI is created in layers of interaction that appeal to the human senses (sight, touch, auditory and more).
- They include both input devices like a keyboard, mouse, trackpad, microphone, touch screen, fingerprint scanner, e-pen and camera, and output devices like monitors, speakers and printers.

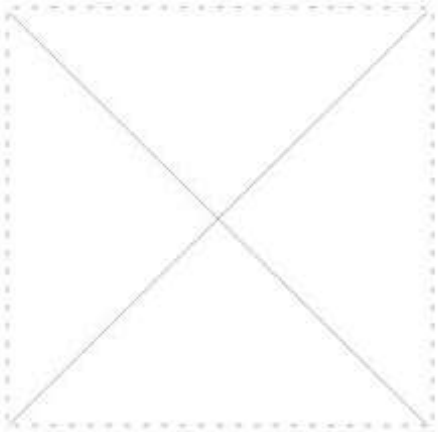
Types of User Interface

- Form-based user interface
- Graphical user interface
- Menu-driven user interface
- Touch user interface (haptics)
- Voice user interface (Eye - free User interface)

UX

User experience is the overall experience a user has with a company's products or services.

Learn more →



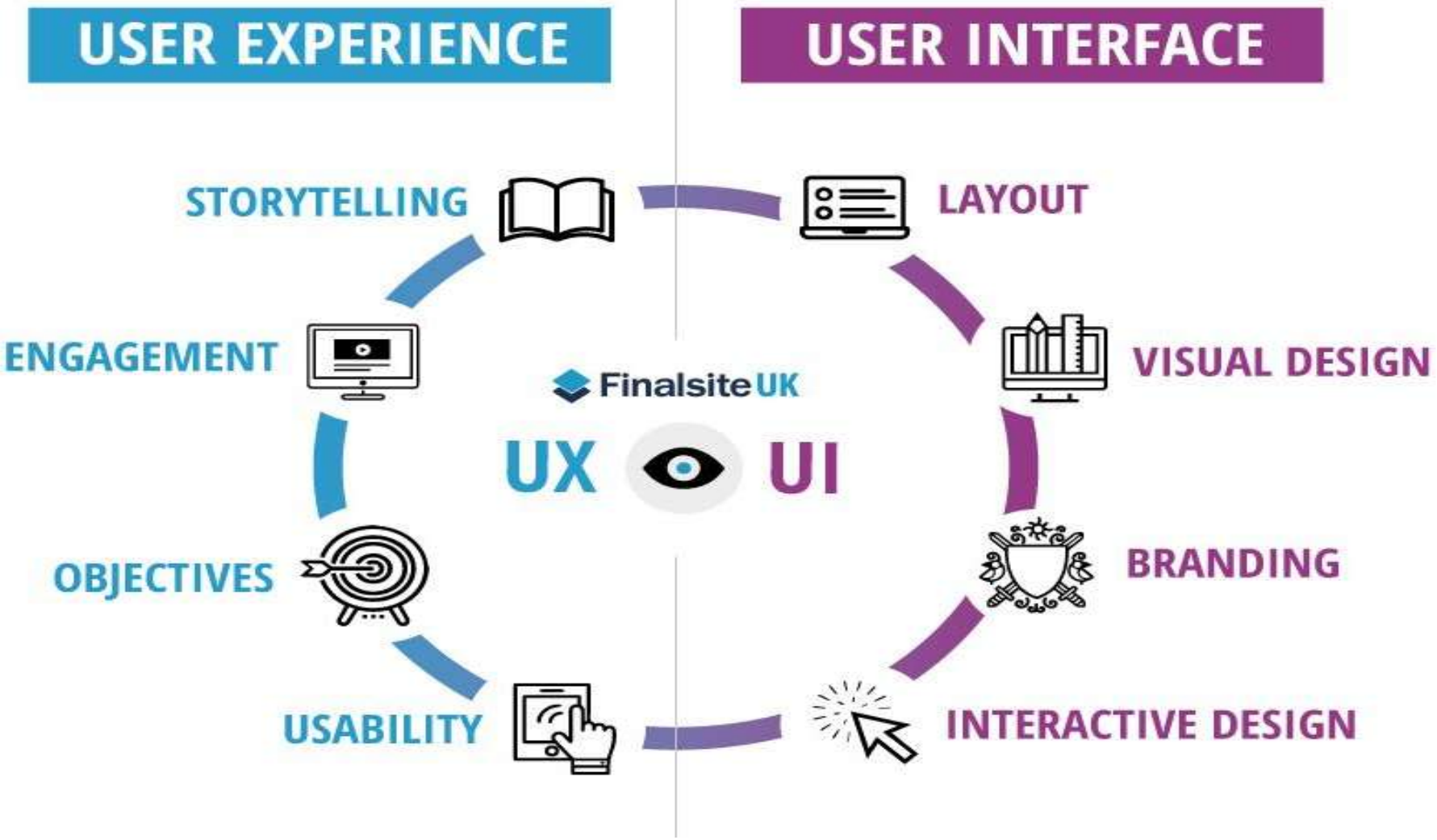
VS

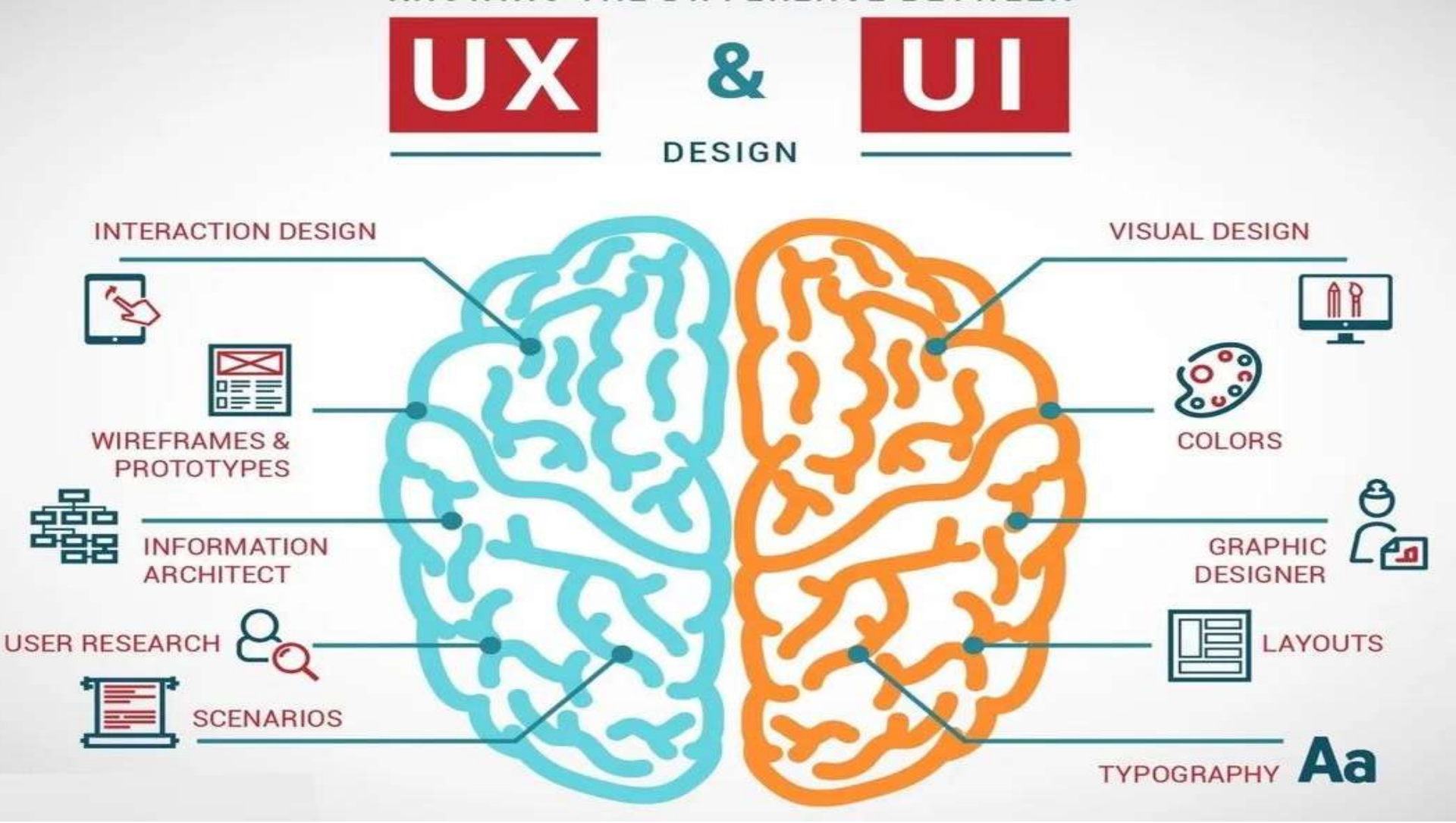
UI

User interface design is focused on the look and layout — how each element of the product will look.

Learn more →




USER EXPERIENCE
USER INTERFACE
STORYTELLING
LAYOUT
ENGAGEMENT
VISUAL DESIGN
Finalsite UK
UX
UI
OBJECTIVES
BRANDING
USABILITY
INTERACTIVE DESIGN


UX
&
UI
DESIGN
INTERACTION DESIGN
VISUAL DESIGN
WIREFRAMES &
PROTOTYPES
COLORS
INFORMATION
ARCHITECT
GRAPHIC
DESIGNER
USER RESEARCH
LAYOUTS
SCENARIOS
TYPOGRAPHY
Aa

UX

HUMAN-FIRST APPROACH TO PRODUCT DESIGN

APPLICATION:
Physical and digital products

FOCUS:
The full experience from a user's first contact to the last

CREATES:
Structural design solutions for pain points that users encounter anywhere along their journey with the product

RESULTS IN:
Products that delight users with their effectiveness

UI

HUMAN-FIRST APPROACH TO DESIGNING THE AESTHETIC EXPERIENCE OF A PRODUCT

APPLICATION:
Digital products only

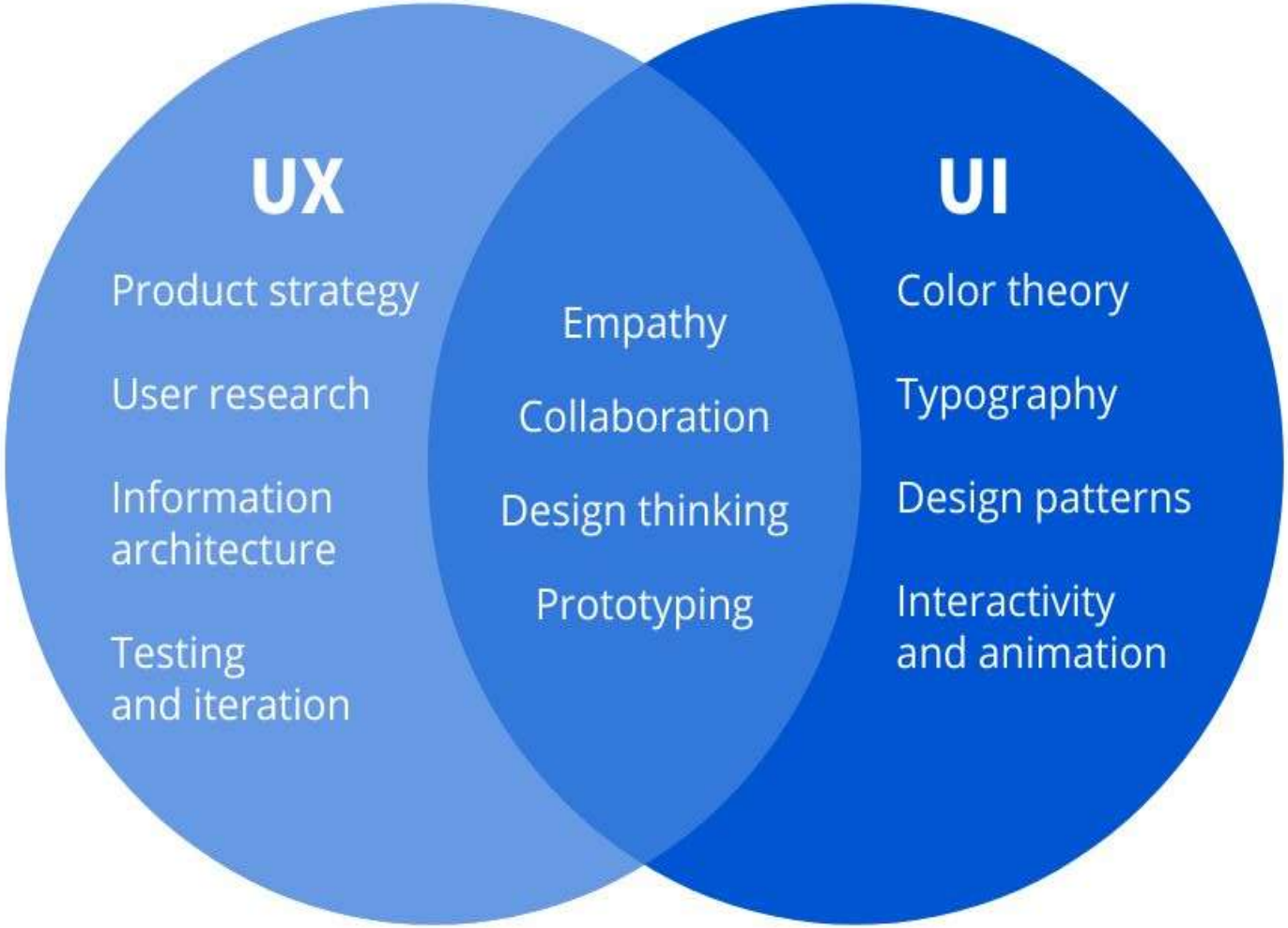
FOCUS:
Visual touchpoints that allow users to interact with a product

CREATES:
Combinations of typography, color palettes, buttons, animations, and imagery

RESULTS IN:
Products that delight users aesthetically

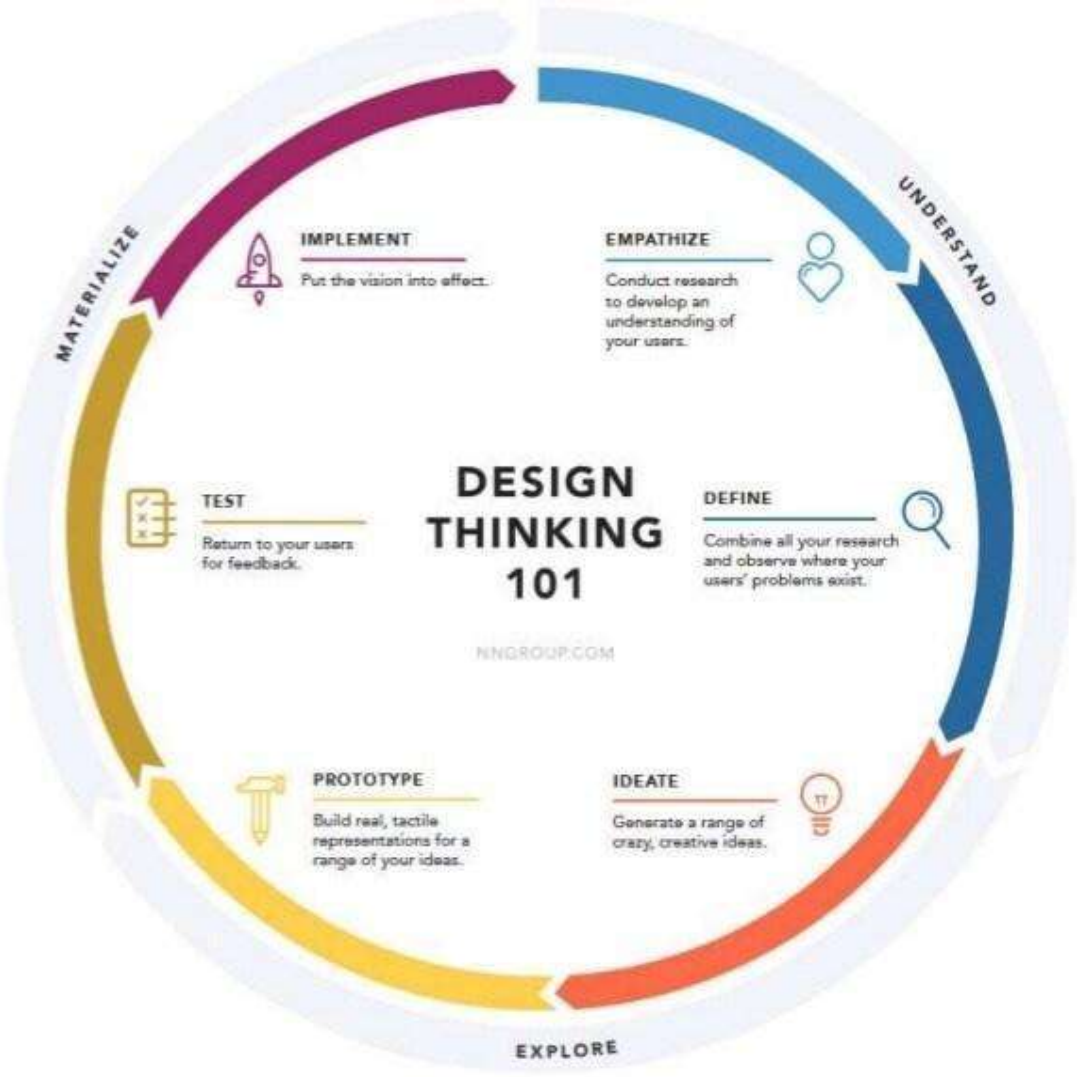
UX vs. UI designers

UX designer	UI designer
Interaction designer	Visual designer
Charts the user pathway	Chooses color and typography
Plans information architecture	Plans visual aesthetic
Expert in wireframes, prototypes, and research	Expert in mockups, graphics, and layouts


UX
Product strategy
User research
Information architecture
Testing and iteration
Empathy
Collaboration
Design thinking
Prototyping
UI
Color theory
Typography
Design patterns
Interactivity and animation

DESIGN THINKING STEPS

a. Empathize
b. Define
c. Ideate
d. Prototype
e. Test
f. Implement



Empathy Map: Student



Learner: MSc Health Sciences student – Full time, thesis

Think

- There are gaps in the program: need more practical experience.
- What's next for graduates?
- Need transparency from faculty members.
- Ask reflective questions.

Feel

- Nervous about making sure I am up to speed with everyone else.
- Enthusiastic about a new model of learning.
- Excited to learn new skills and apply them to my day-to-day.
- Feel disconnected from other students.

Say

- Thesis was challenging from a distance.
- Courses look very interesting and very relevant.
- Thesis process: Need more clarity and guidance.
- To do research in education, you need to do a supervised project.

Do

- Reading and posting in discussion forums.
- Connect with recent graduates from the program.
- Design and draft ways to connect CHSE and HSEDUC.
- Organize my schedule to slot in dedicated time for reading and posting.

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User Experience Design (UxD)

- User experience design is about understanding customers' feelings when interacting with a website, app, or product.
- A good UX design is functional, desirable, accessible, credible, findable, usable, and valuable.
- User experience does not depend only on how the product is designed. There are other aspects as well that make UX design more challenging.
- This includes the user and the situation where they use the product.

Importance of User Experience Design

- User interface is important to meet user expectations and support the effective functionality of your site.
- A well-executed user interface facilitates effective interaction between the user and the program, app or machine through contrasting visuals, clean design and responsiveness.
- When designing a UI for your site, it's important to consider the user's expectations in terms of accessibility, visual aesthetic and ease of use.
- An optimal mix of effective visuals and efficient responsiveness will improve your site's conversion rates, as it anticipates the needs of the user and then satisfies those needs.

Important Elements of Great UI

- **Information architecture:**
  - The functionality of a site is built according to the IA.
  - Structuring and organizing your website's content in a logical way is important to help users navigate the site with minimal effort.
  - Components of IA include three main types of organizational structures:
    - hierarchical (level of importance),
    - sequential (logical order of steps) and
    - matrix (in which the user chooses the organization of the content they see).
- **Example:** Navigation elements (buttons, tabs, icons), labels (terminology), search features (search bar) and organization systems (categories)

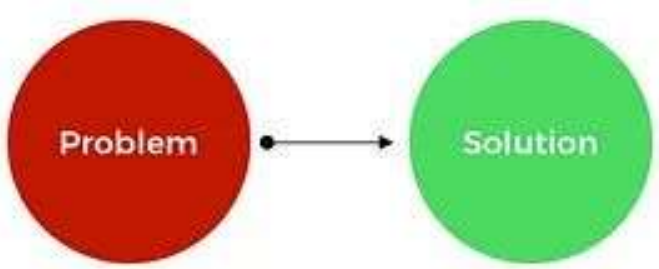
Important Elements of Great UI

- **Interactive design:**
  - Elements of ID aim to turn passive readers into active participants by presenting instances of user input.
  - Keeping the user in mind while creating the UI will help to improve the interactivity and execution of specific behaviors that satisfy user needs.
  - Additionally, efficiently-designed interactive UIs can 'learn' to anticipate and remedy any problems that may come up before they negatively impact the user's experience**.**
- **Example: Social-share features, toggles, buttons.**

Important Elements of Great UI

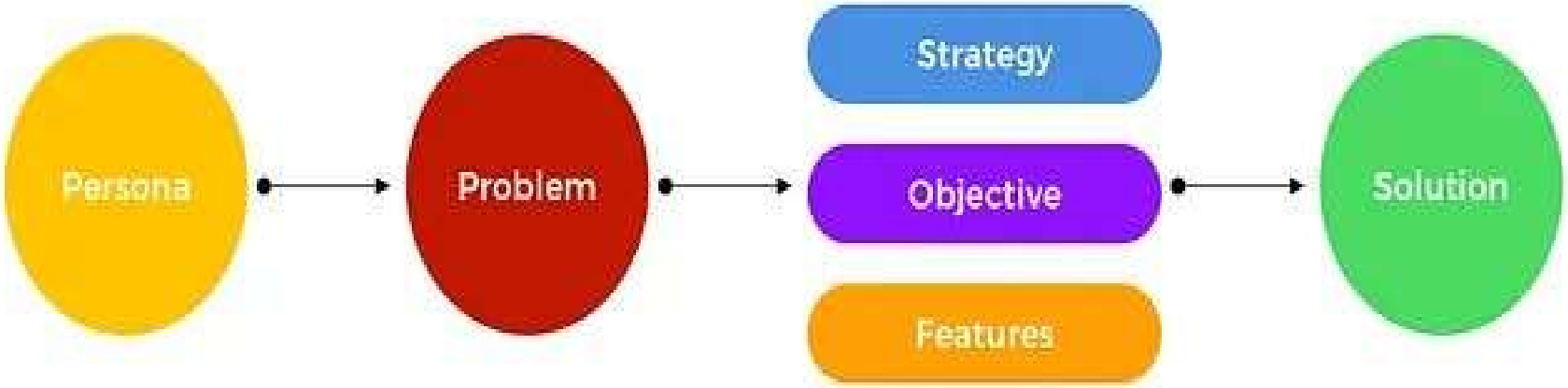
- **Visual design:**
  - The importance of your site's aesthetic value cannot be underestimated.
  - Effective design utilizes color, contrast, font, video and photo elements to attract visitors and make it easy for them to read and works with the content, rather than around it, to create a logical, intuitive flow of functionality.
- Example: Contrast, color, white space, typography, mobile optimization.

Importance of User Experience Design



✕ Don't jump straight to solutions

Importance of User Experience Design



✓ Take time to properly define the persona, problem, strategy, objective and features will provide more effective solution

Persona

Who has the problem?

Problem

What is the problem?

Strategy

How will we solve the problem?

Objective

What will the solution achieve?

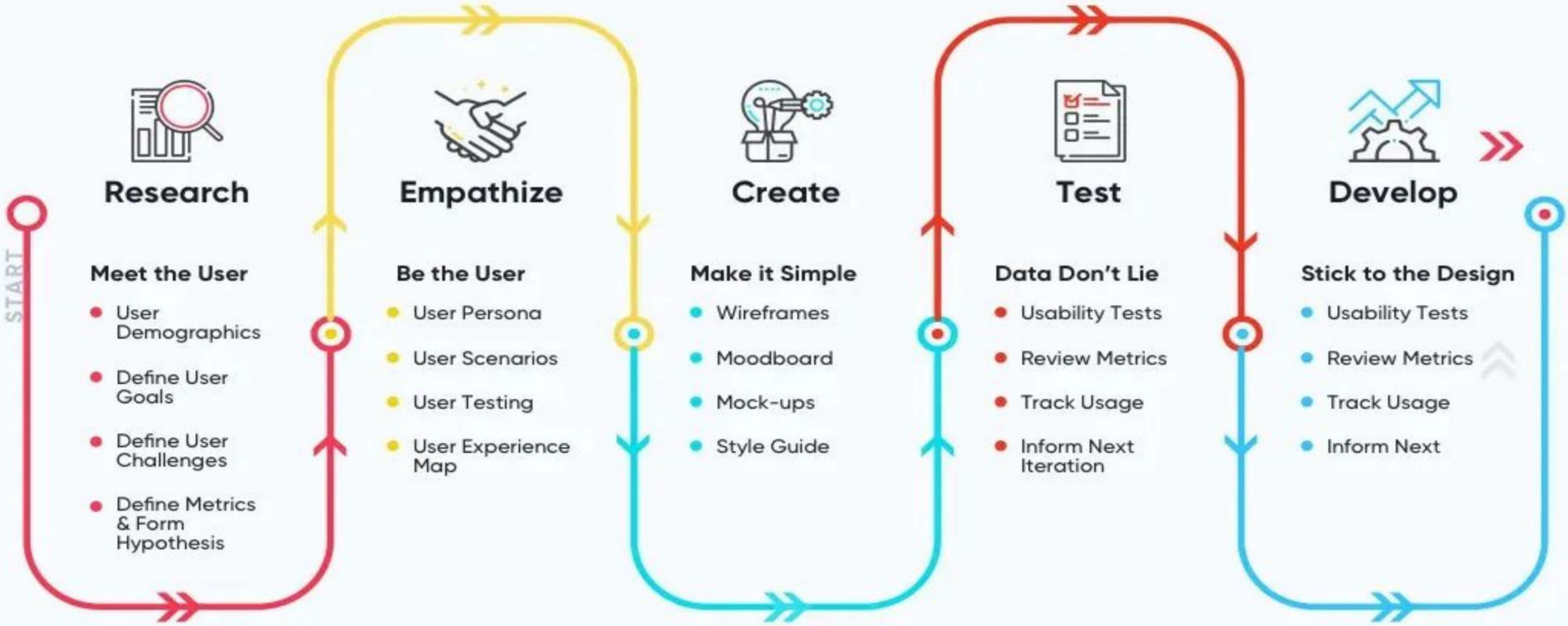
Features

What features are required to accomplish the objective?

Solution

What will the product look like and how will it function?

User experience design process



net solutions

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Usability

- Usability is “extent to which a system, product or service can be used by specified users to achieve specified goals with effectiveness, efficiency and satisfaction in a specified context of use”
  - What do they (the users) need?
  - What are their capabilities?
  - What are their limitations?

- Usability is a very **objective and measurable** goal because it has everything to do with performance.

Usability



Carelman pot

The impossible teapot. (Author's collection,after Carelman's "Coffeepot for Masochists.")

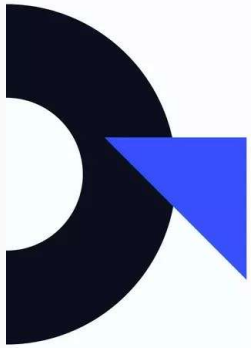




Google and Yahoo

Provides examples of good usability websites

Usability



"Usability is about human behavior. It recognizes that humans are lazy, get emotional, are not interested in putting a lot of effort into, say, getting a credit card and generally prefer things that are easy to do vs. those that are hard to do."

— DAVID MCQUILLEN, EX-SWISS BANKER AND FOUNDER OF SUFFERFEST CYCLING WORKOUT RESOURCES

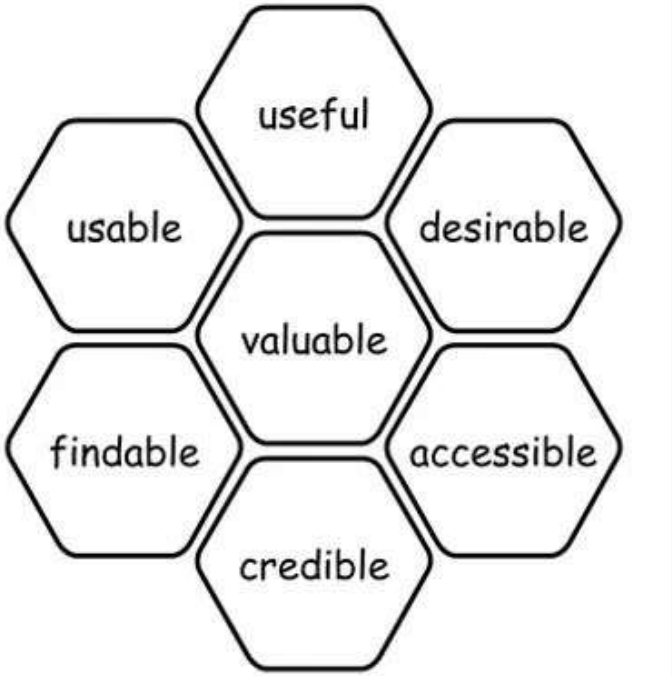
Characteristics of Usable Product

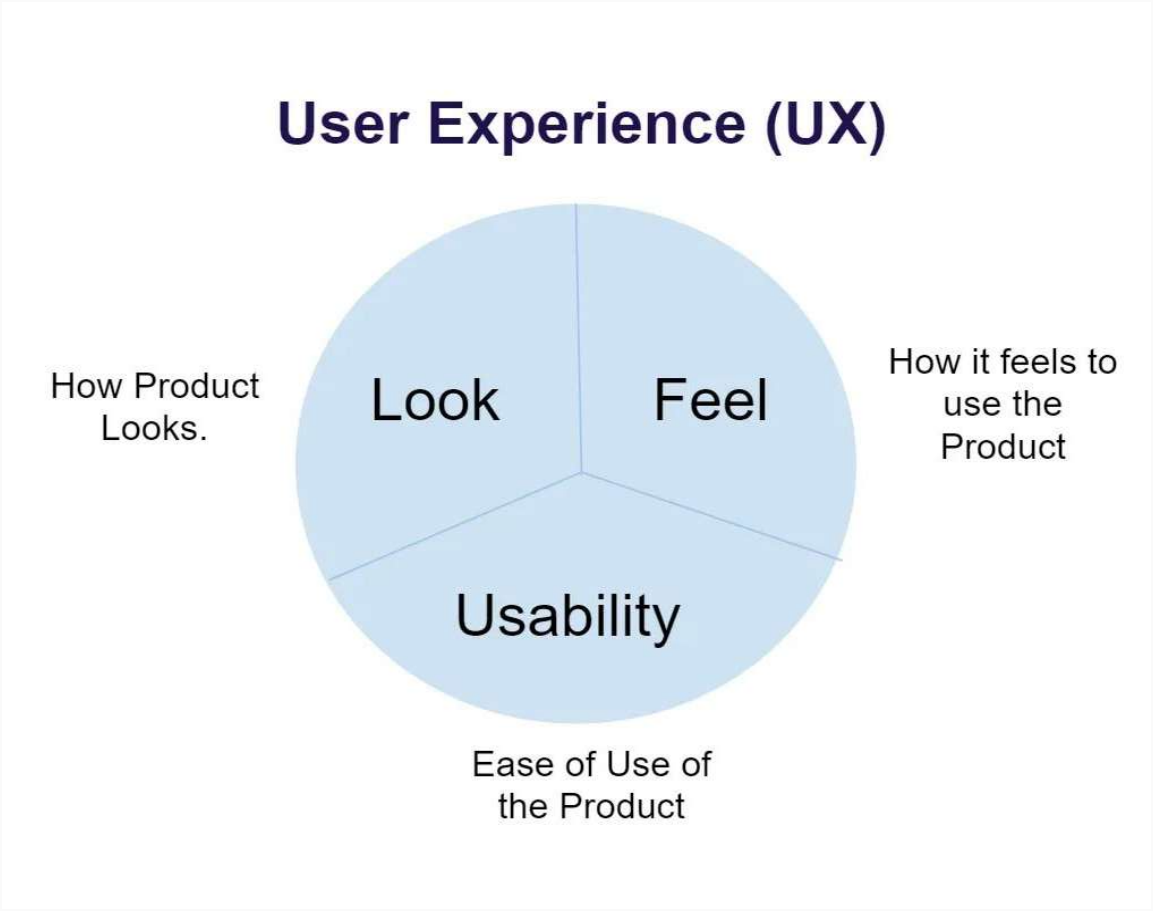
- Is the product easy to use?
- Is it easy to learn?
- Is it intuitive?
- Does my product promote efficiency for the user?
- Does it prevent errors and mistakes?

User Experience

- UX is a large concept that encapsulates all the “user’s perceptions and responses that result from the use and/or anticipated use of a system, product or service”
- It take into account the technical facet, aesthetics, and brand association, but also more subjective and individual variables, like the users' expectations, previous experiences, and the context in which the product is being used.

Usability and User Experience





Usability vs User Experience (UX)

USABILITY

Effectivness
Efficiency
Learnability
Error Prevention
Memorability

VS

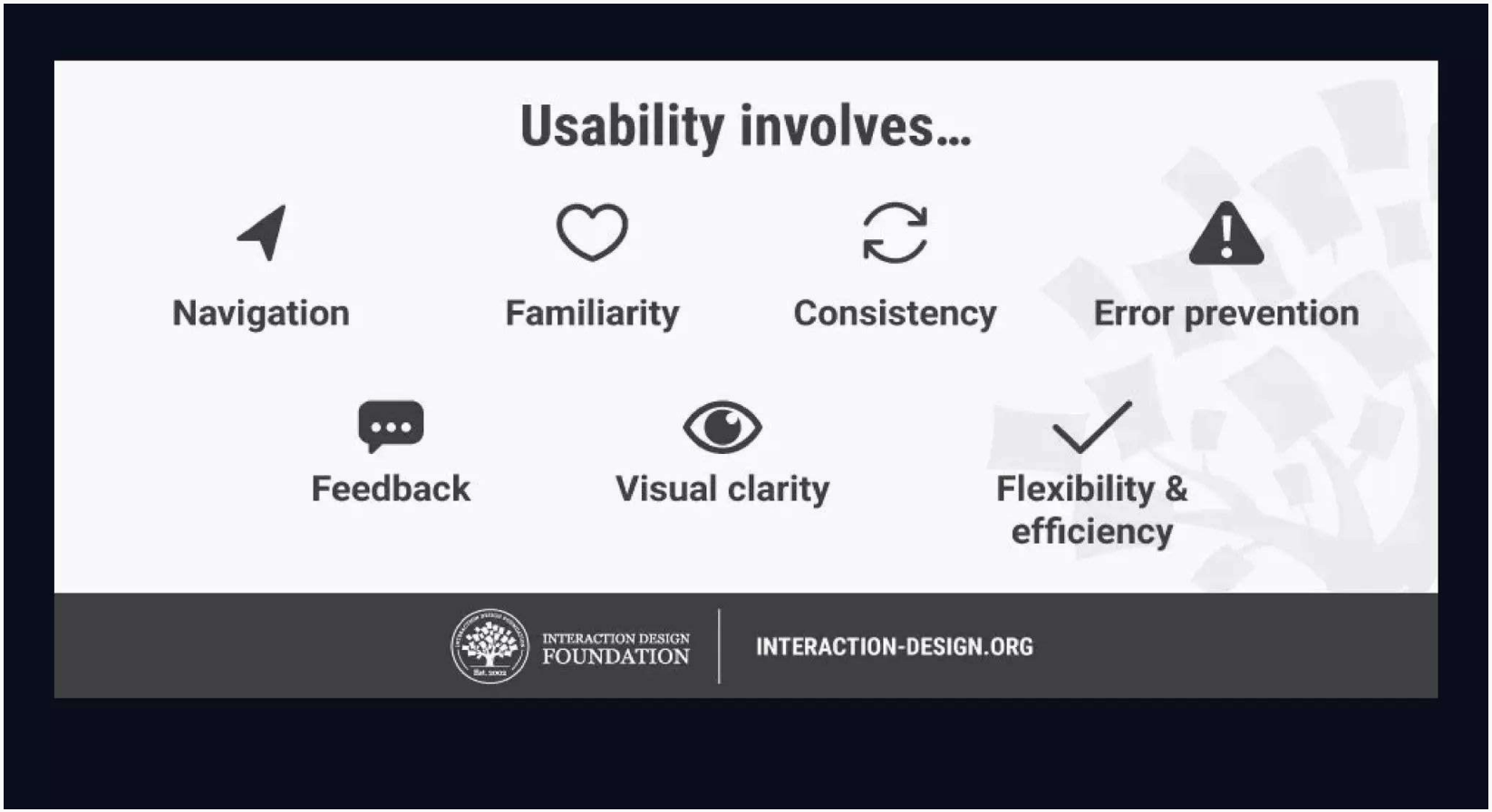
USER EXPERIENCE

Satisfcation
Enjoyment
Pleasure
Fun
Value



userpilot.com

Usability


Usability involves...
Navigation
Familiarity
Consistency
Error prevention
Feedback
Visual clarity
Flexibility & efficiency
INTERACTION DESIGN FOUNDATION
INTERACTION-DESIGN.ORG

Fifteen years ago, Companies:

Competed on : Price

Today it's : Quality

Tomorrow it's : Designs

When companies are competing at equal price & functionality Design is the only differential that matters"

– Mark Dziersk,
quoted in TIME Magazine

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“Design is not just what it looks like and feels like.
Design is how it works.”
-Steve Jobs